

Discovering Latent Musical Structure Using Spotify Audio Features

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Abstract—Digital music platforms generate large-scale datasets describing the acoustic properties of songs. While genre labels and popularity metrics provide surface-level categorization, they often fail to capture deeper structural relationships between tracks. This project proposes a data mining approach to discover latent musical structures using Spotify audio features. By applying dimensionality reduction, clustering, and anomaly detection techniques, the project aims to uncover meaningful and interpretable patterns that describe how songs relate to one another beyond predefined labels.

I. INTRODUCTION

The increasing availability of large-scale music datasets has created new opportunities for discovering patterns in musical content. Streaming platforms such as Spotify extract quantitative audio features that describe rhythm, intensity, and emotional tone.

The objective of this project is to apply data mining techniques to identify latent structure in music using Spotify audio feature data. Unlike traditional machine learning approaches focused on prediction, this project emphasizes **pattern discovery and interpretation**. The goal is to understand how songs relate to one another based on acoustic similarity.

By clustering songs based on audio features and identifying anomalous tracks, this study aims to uncover interpretable musical groupings that reflect underlying sound characteristics.

II. DATASET DESCRIPTION

This project uses the **Spotify Tracks Dataset**, publicly available on Kaggle:

<https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset>

The dataset contains:

- Approximately 114,000 tracks
- 21 features
- Metadata and numerical audio attributes

A. Key Features

The dataset includes several numerical audio features:

- **Danceability (0–1)**: Measures how suitable a track is for dancing based on rhythm and tempo
- **Energy (0–1)**: Represents intensity and activity level
- **Loudness (dB)**: Overall volume of the track
- **Acousticness (0–1)**: Likelihood that a track is acoustic
- **Valence (0–1)**: Musical positivity (happy vs sad)
- **Tempo (BPM)**: Speed of the song

Additional features include instrumentalness, speechiness, and liveness, along with metadata such as popularity and explicit content.

These features are computed using Spotify’s internal audio analysis algorithms and represent measurable properties of sound such as rhythm, intensity, and mood. Because the dataset is primarily numerical, it is well-suited for clustering and dimensionality reduction techniques.

B. Expected Preprocessing

Preprocessing will include:

- Removing non-informative identifiers (track names, IDs)
- Handling missing values in metadata
- Normalizing feature scales
- Selecting relevant numerical attributes for analysis

III. DISCOVERY QUESTIONS

This project is guided by the following discovery-oriented questions:

- Are there natural clusters of songs based on their acoustic features such as energy, tempo, and acousticness?
- What patterns exist between discovered clusters and attributes such as popularity and explicit content?
- Are there anomalous tracks whose audio profiles deviate significantly from typical musical patterns?

These questions focus on discovering structure and relationships within the data rather than predicting outcomes.

IV. PLANNED TECHNIQUES

The proposed data mining pipeline consists of multiple stages:

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Raw Data
↓
Preprocessing & Feature Selection
↓
Normalization
↓
Dimensionality Reduction (PCA)
↓
Clustering (K-Means, DBSCAN)
↓
Anomaly Detection
↓
Interpretation & Evaluation
```

A. Dimensionality Reduction

Principal Component Analysis (PCA) will be used to reduce dimensionality and visualize latent structure. PCA helps identify dominant patterns and reduces redundancy among correlated features.

B. Clustering

Two clustering approaches will be applied:

- **K-Means:** Identifies global structure by partitioning data into clusters. The number of clusters (k) will be determined using the elbow method and silhouette score.
- **DBSCAN:** Detects density-based clusters and identifies noise points. This method captures irregular cluster shapes that K-Means may miss.

C. Anomaly Detection

Anomaly detection techniques (e.g., Local Outlier Factor) will be used to identify songs that deviate from typical patterns. These outliers may represent unique or experimental tracks.

D. Evaluation and Interpretation

Cluster quality will be evaluated using:

- Silhouette Score
- Davies-Bouldin Index

Results will be interpreted in terms of musical characteristics to provide meaningful insights.

V. PRELIMINARY TIMELINE

- **M2:** Data preprocessing, exploratory data analysis, and initial clustering
- **M3:** Implementation of clustering, DBSCAN, and anomaly detection
- **M4:** Final interpretation, validation, and report preparation

A. Anticipated Challenges

Potential challenges include:

- Selecting optimal clustering parameters (e.g., k , epsilon)
- Ensuring clusters represent meaningful musical structure
- Interpreting overlapping clusters due to continuous feature distributions

VI. CONCLUSION

This proposal outlines a data mining approach to uncover latent musical structures using Spotify audio features. By focusing on discovery rather than prediction, the project aims to reveal interpretable patterns that enhance understanding of musical similarity and diversity in large-scale datasets.